

LECTURE 2 — ENERGY GENERATION

PHY199-RAP · Energy in the 21st Century

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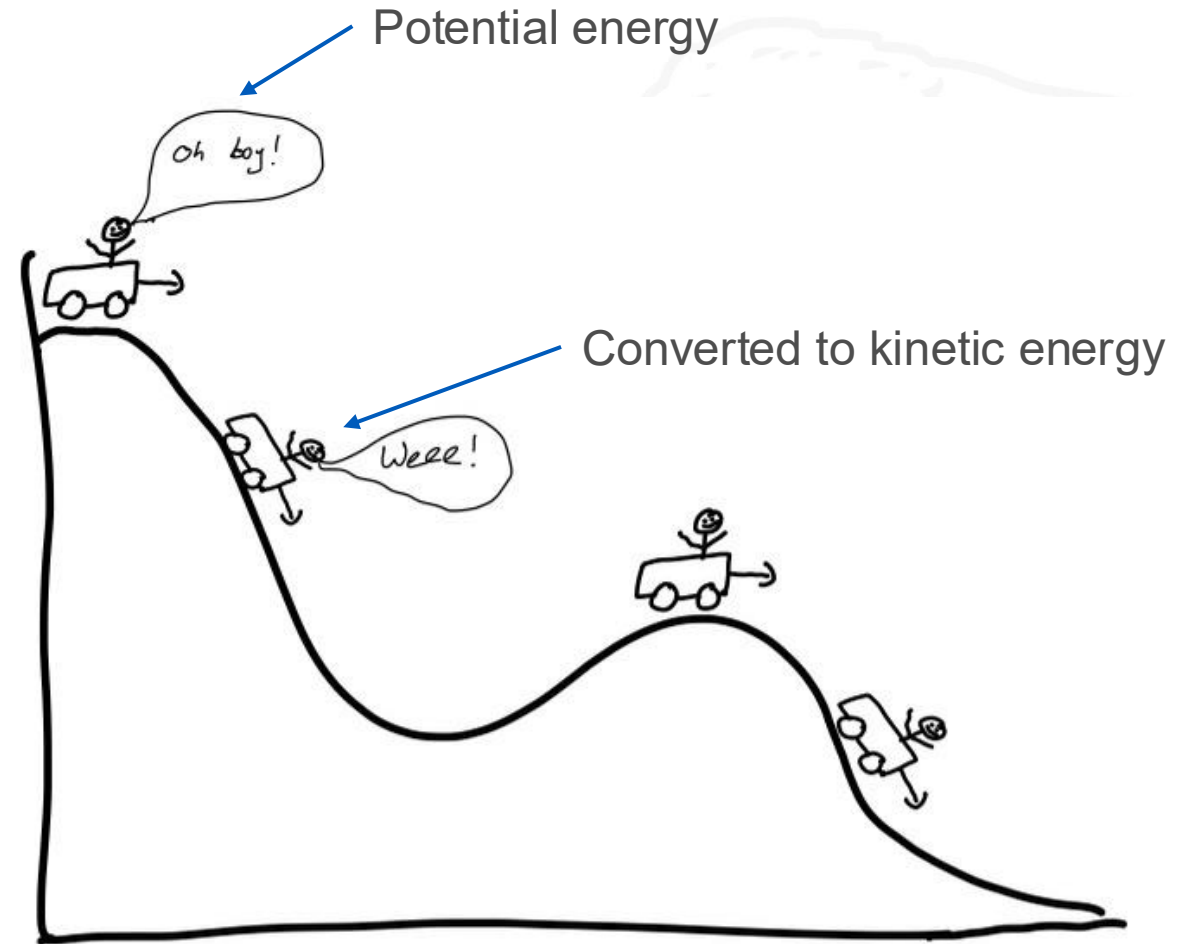


FORCES AND ENERGY



Energy

- **Two types** of energy:
 - **Kinetic** : something in motion
 - Moving cars, people, electrons, planets, quarks, etc
 - **Potential** : something that can be stored to convert into motion later
 - Electric potential energy, gravitational potential energy
- This class is about **energy generation**, i.e. converting potential energy from something into kinetic energy of something else (primarily **machinery, electronics, or vehicles**)

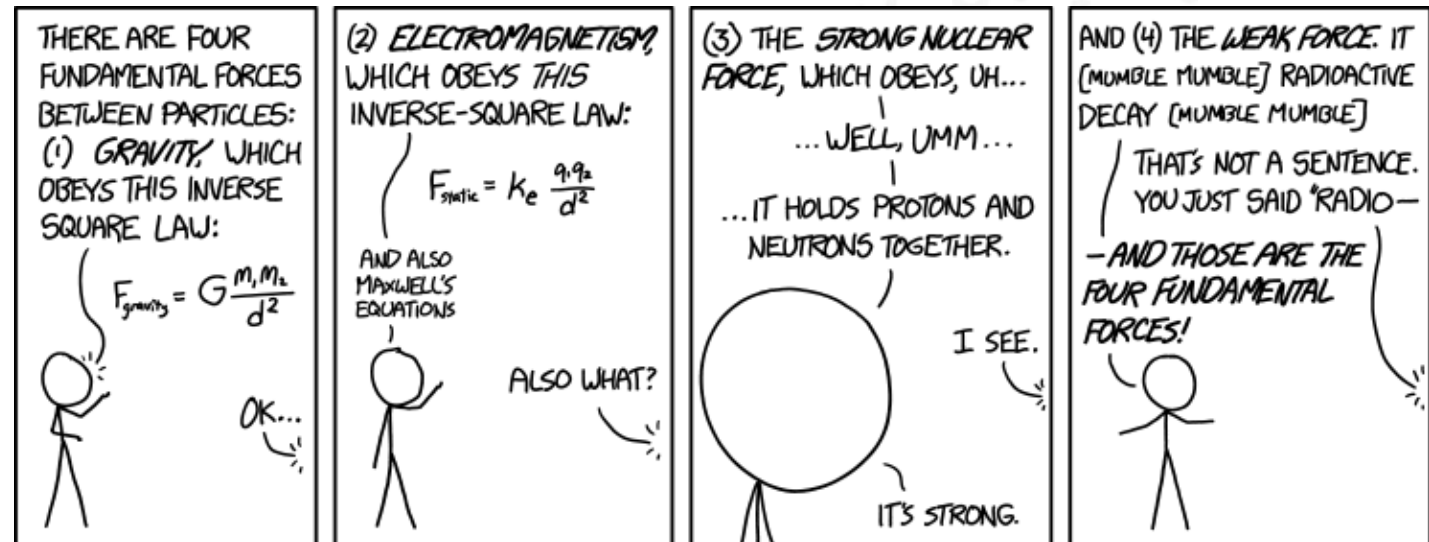


I wish I could blame AI, but I'm just bad at drawing

The Four Forces

- Gravity
- Electromagnetism
- Strong nuclear interaction
- Weak nuclear interaction

All energy comes from one of these four sources. We'll investigate them all.



Of these four forces, there's one we don't really understand."

"Is it the weak force or the strong--"

"It's gravity." <https://xkcd.com/1489/>

Force

- Forces are what act upon bodies to change their acceleration (or shape, or pressure of fluids). In 1 dimension:

Force:
mass * distance / time²
(1 Newton = 1 kg m/s²)

$$F = ma$$

mass (kg)

Acceleration:
distance / time² (m/s²)

- (Caveat: Force and acceleration are VECTORS. We won't go into it though)
- Velocity is how your position changes per unit time (20 miles per hour, 6 meters per second)
- Acceleration is how your velocity changes per unit time (9.8 meters per second per second, or 9.8 m/s²)

Work

- Applying a FORCE over a DISTANCE means you did WORK

Work:
mass * distance² / time² (1
Joule = 1 kg m²/s²)

$$W = F \cdot d$$

Force:
mass * distance / time²
(1 Newton = 1 kg m/s²)

distance (m)

- Don't panic this is going to get easier in a minute, bear with me.

Not responsible for this, just for enrichment

Work and Kinetic Energy

- Work is the thing that changes KINETIC energy

$$W = F \cdot d = \Delta KE = \frac{1}{2} m \Delta v^2$$

- So we see:

Energy:
1 Joule = 1 kg m²/s²

$$KE = \frac{1}{2} m v^2$$

mass (kg)

Velocity:
distance/time
(m/s)

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For those who love calculus like me:

$$F = ma = m \frac{dv}{dt}$$

$$W = \int F dx$$

$$W = \int m \frac{dv}{dt} dx$$

$$W = \int m \frac{dx}{dt} dv$$

$$W = \int m v dv$$

$$W = \frac{1}{2} m \Delta v^2$$

Kinetic and Potential Energy

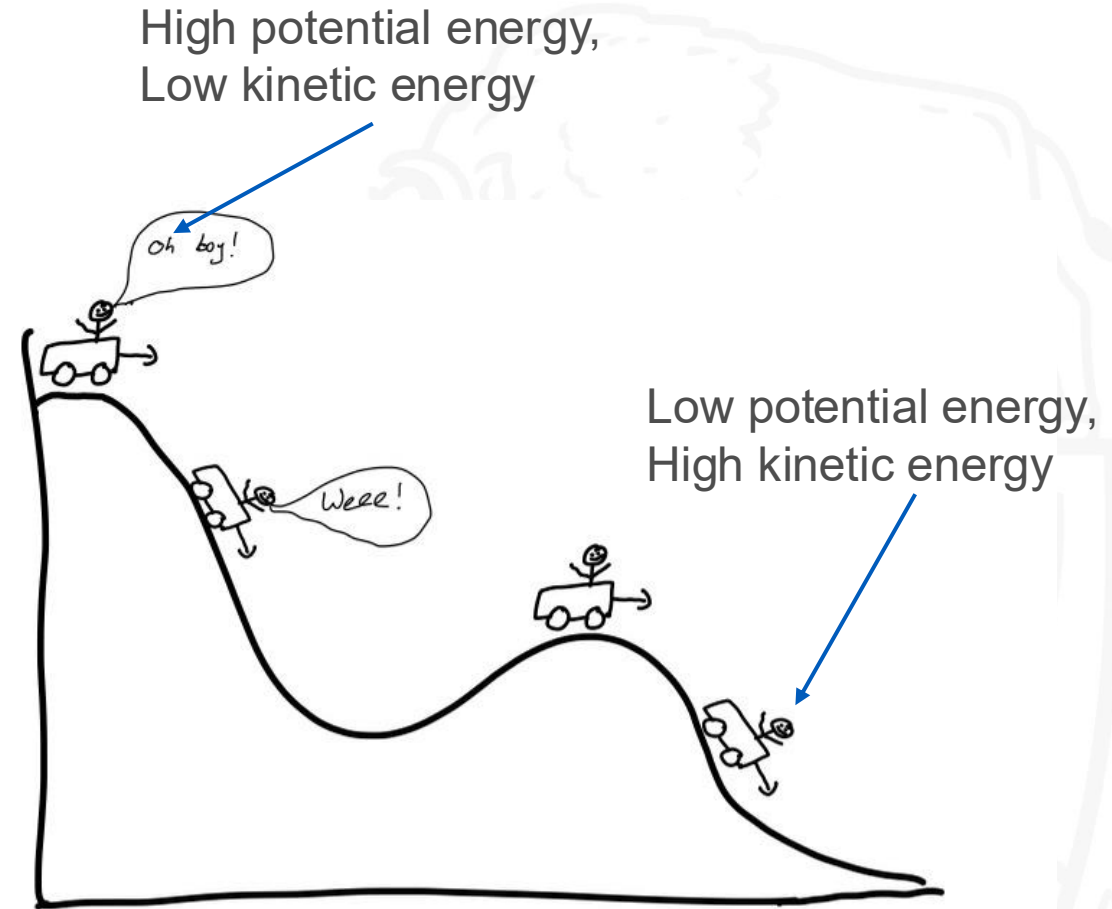
- If you can put energy INTO something, sometimes it can be retrieved later (“**conservative forces**”)
 - Like gravity: What comes up, must come down
- For a conservative force, the total energy (kinetic plus potential) remains the same!

$$\text{Energy} = \Delta KE + \Delta PE = \text{constant}$$

- Still equal to the force times the distance
 - For gravity near the earth’s surface:

$$PE = mg \cdot h$$

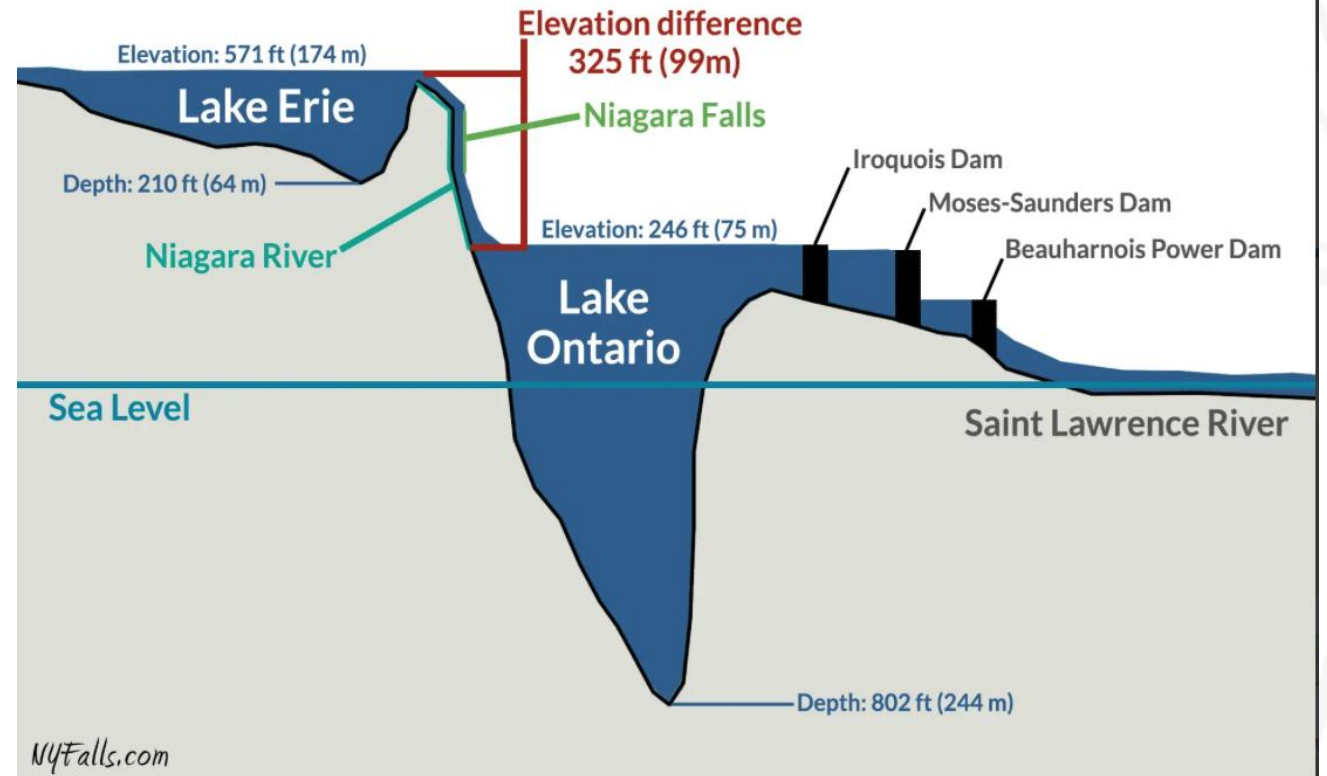
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Using Energy

- Example: water running in a river or down a huge waterfall
 - Niagara Falls!

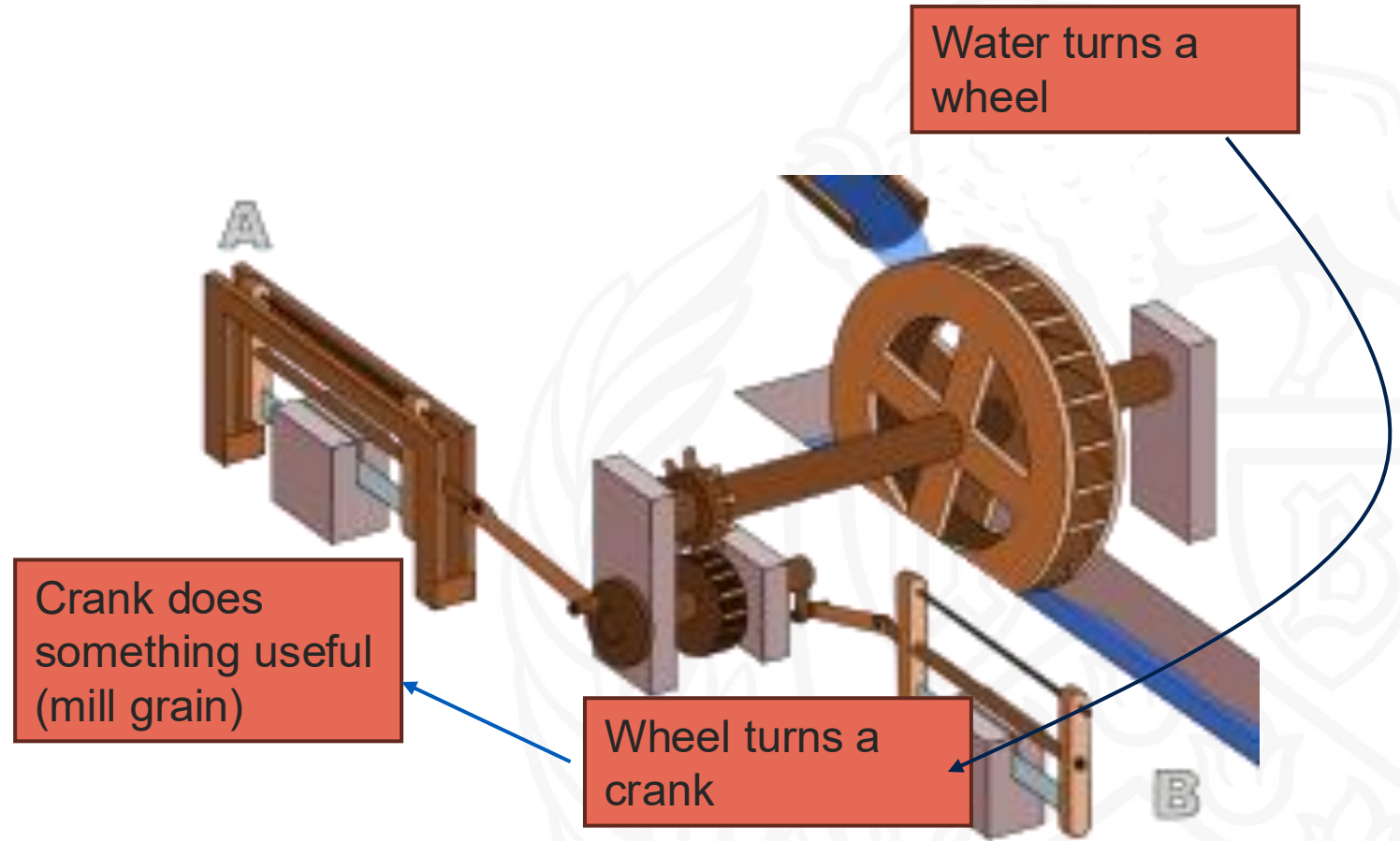
You ARE responsible to understand that Niagara Falls (and all hydroelectric power) is powered ultimately by GRAVITY



<https://nyfalls.com/niagara-falls/faq-6/>

Using Energy

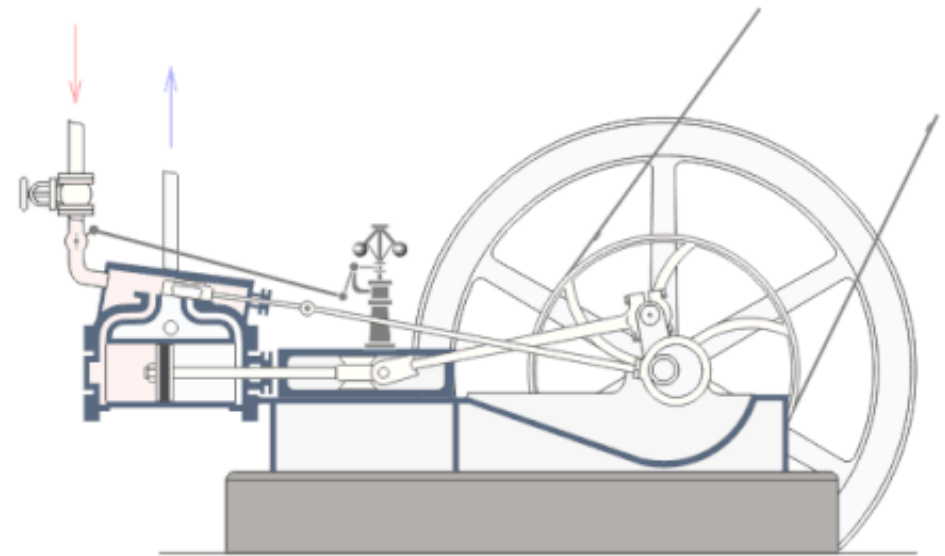
- Running rivers have been used for centuries to convert the gravitational energy of the earth to mechanical energy
- **Water mills!**



https://en.wikipedia.org/wiki/Water_wheel#/media/File:Römische_Sägemühle.svg

Kinetic and Potential

- The industrial revolution replaced moving water (from **gravity**) to burning coal (**thermal energy from chemical energy**) and using STEAM instead of water
- Critically: This is **portable!**



https://en.wikipedia.org/wiki/Steam_engine#/media/File:Steam_engine_in_action.gif

Power

You ARE responsible to understand that power is the energy expended per unit time

- Power is defined as the amount of **energy expended per unit time**

Power:
1 Watt = 1 kg m²/s³

$$P = \frac{W}{\Delta t}$$

Energy:
1 Joule = 1 kg m²/s²

time (s)

- We often go the other way around and say that Energy is the Power times the time (like kilowatt-hours, which is what **goes on your electricity bill**)

Energy:
1 kilowatt-hour = 1000 Watts (J/s) * 3600 seconds =
3,600,000 Joules

My electric bill history

Enrollment Information

To enroll with a supplier or change to another supplier, you will need the following information about your account:

Loadzone Frontier

Acct No: 05963-18400 **Cycle:** 17, RAPP

Electric Usage History

Month	kWh	Month	kWh
Aug 25	225	Mar 26	220
Sep 25	184	Apr 26	168
Oct 25	137	May 26	151
Nov 25	160	Jun 26	324
Dec 25	261	Jul 26	587
Jan 26	237	Aug 26	535
Feb 26	238		

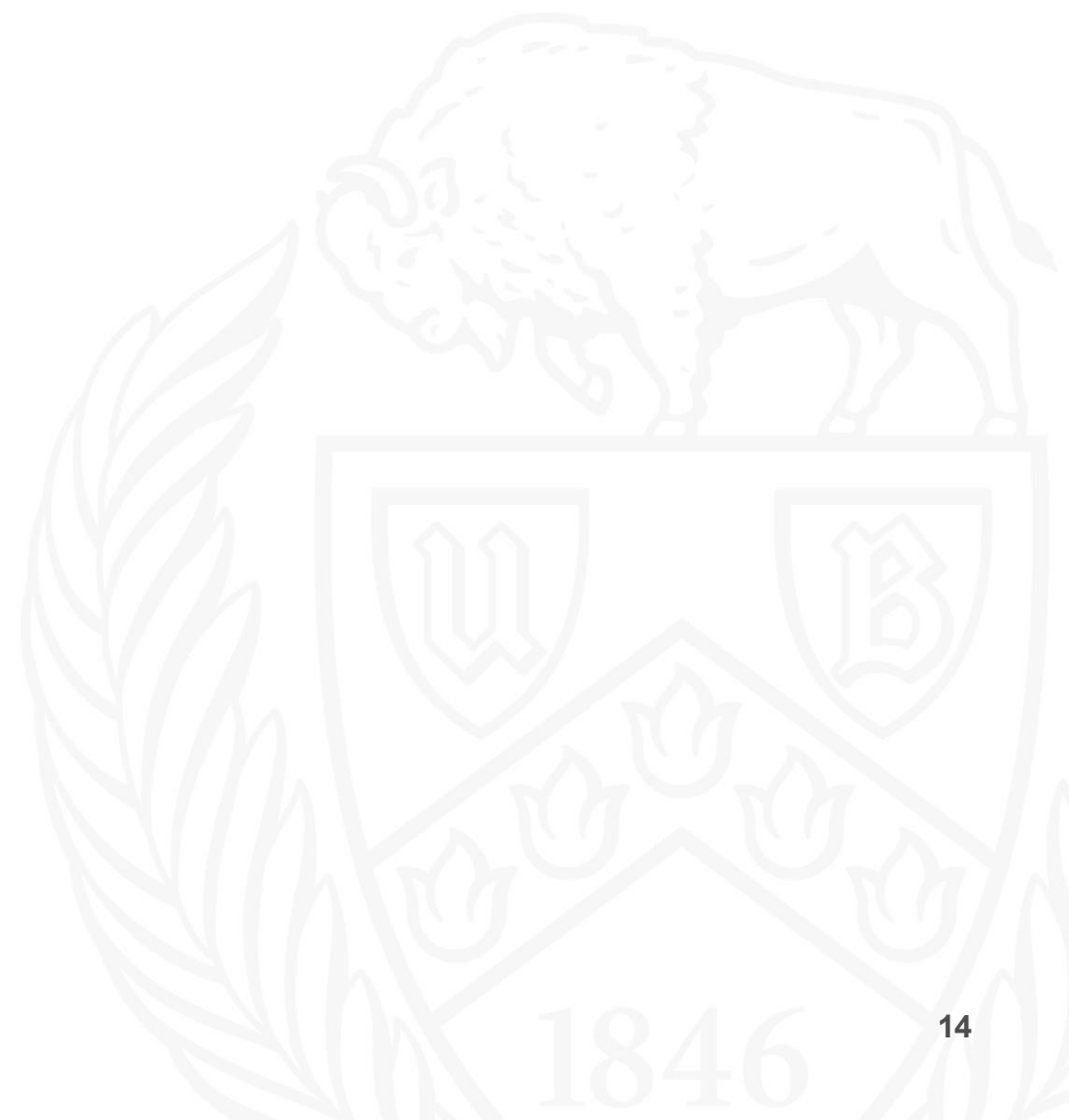
Power

- If the force being used to generate the work is constant, we remember

$$P = \frac{W}{\Delta t} = \frac{F \cdot \Delta d}{\Delta t} = F \cdot v$$

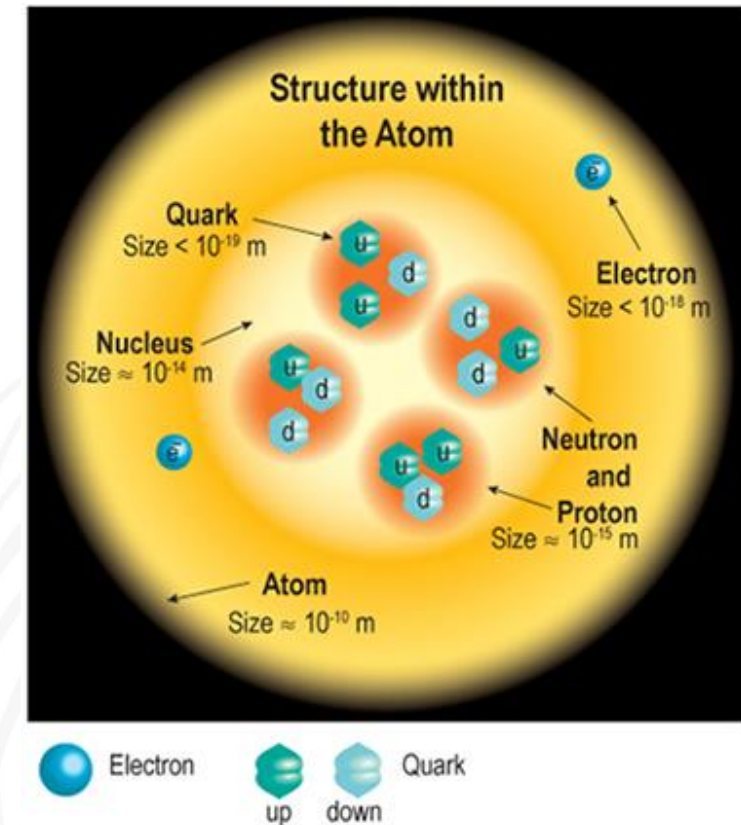
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ELECTRICITY



Electricity

- **Charge** is a fundamental property of matter
 - Based on charge of electrons and protons (one charge = e)
 - Electrons: $-1e$
 - Quarks: $\frac{2}{3}e, \frac{-1}{3}e$
 - Protons are two up quarks and one down quark, EXACTLY $+1e$
 - No idea exactly why
 - Electricity was discovered before the electron, so we use a unit called a **COULOMB** for charge:



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$$1 \text{ Coulomb (C)} = 6.24 \times 10^{18} e$$

Electricity

- Electricity exerts a FORCE on charged things with magnitude:

Force:
mass * distance / time²
(1 Newton = 1 kg m/s²)

$$F = \frac{k q_1 q_2}{r^2}$$

Charge:
Coulombs (C)

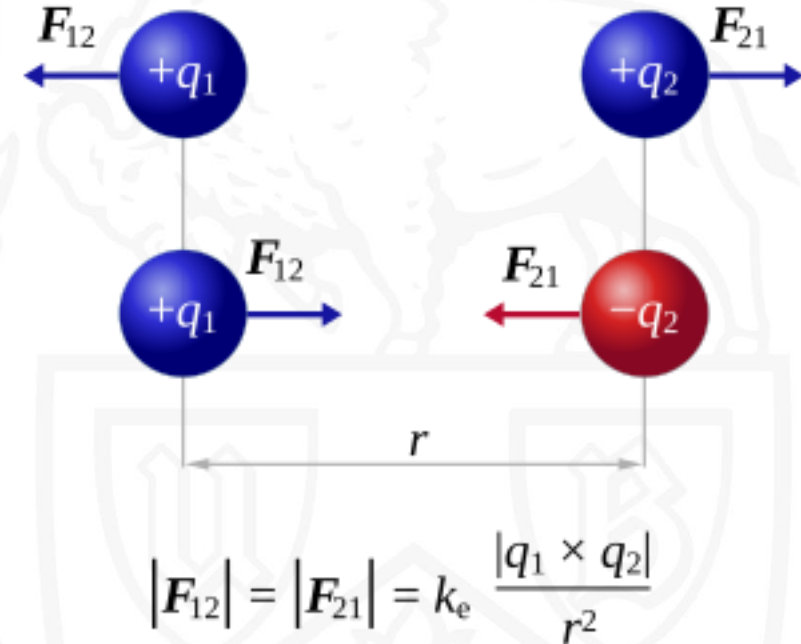
distance (m)

- The constant k depends on the units you use.
- For SI units:

The constant k:
mass * distance³ / charge² * time²

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ew
that's
confusing



https://en.wikipedia.org/wiki/Coulomb%27s_law

Electricity

You ARE responsible to understand that current is the number of charges per unit time

- Practically speaking, for most things we don't actually care about the individual charge, we care about how much charge is **passing through** (the **current**)

$$i = \frac{\Delta Q}{\Delta t}$$

Charge:
Coulombs

Current:
Ampere = Coulombs / second

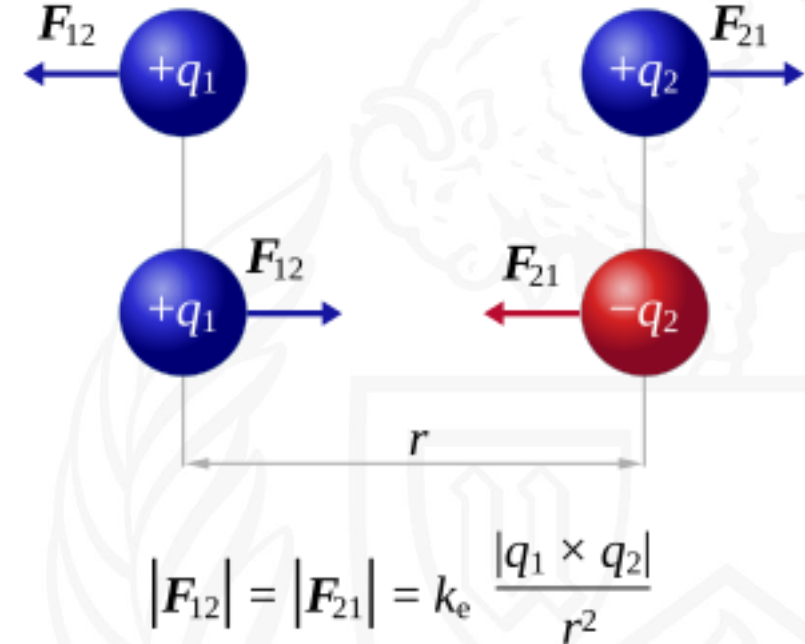
time:
seconds

Electricity

- What about energy and work?

$$F = \frac{kq_1q_2}{r^2}$$

$$W = F \cdot r = \frac{kq_1q_2}{r}$$



Work is the product of the amount of force over the distance

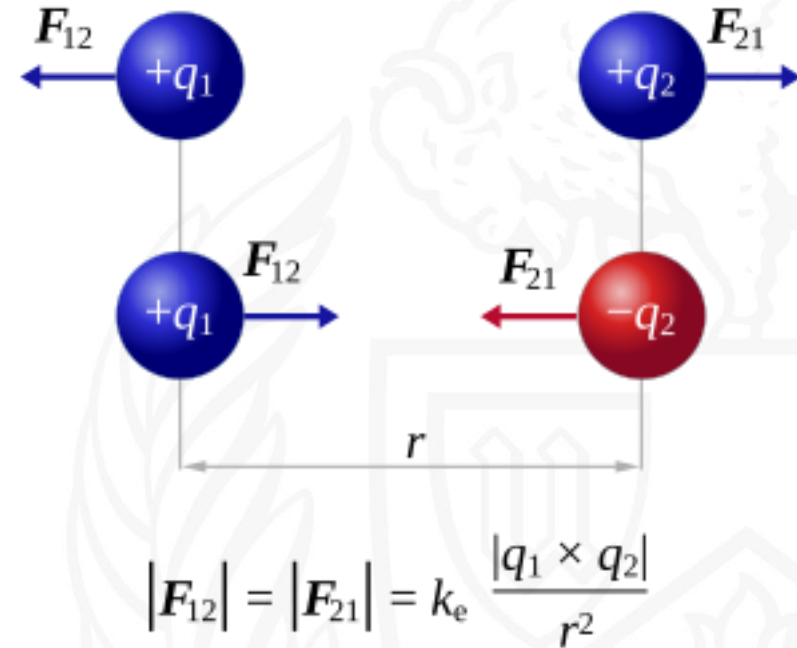
Electricity

- This is hard... let's simplify.
- Develop the concept of an electric FIELD by dividing out one charge
 - The “test charge”
- Then we have the FIELD from the other charge

$$\text{Electric field} = \frac{F}{q_1} = \frac{kq_2}{r^2}$$

Electric field:
Newtons / Coulomb

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https://en.wikipedia.org/wiki/Coulomb%27s_law

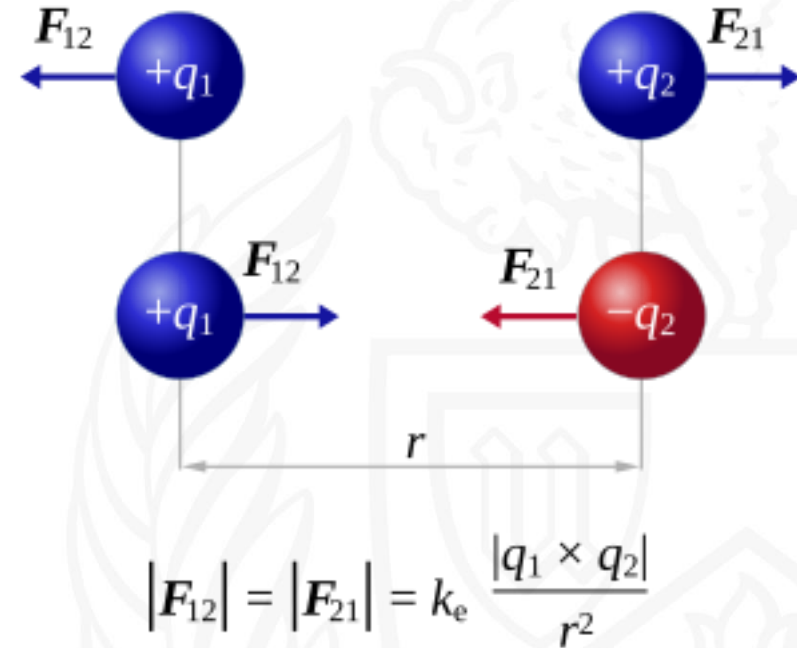
Electricity

- We can also develop the electric **POTENTIAL** from the electric **ENERGY (or work)** in the same way

$$V = \frac{PE}{q_1} = \frac{kq_2}{r}$$

Electric potential:
Joules / Coulomb or **VOLTS**

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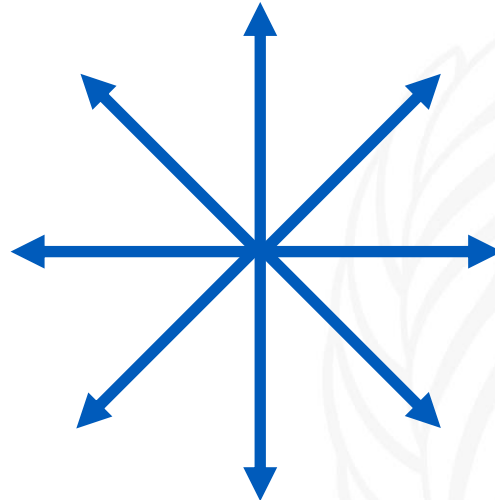
https://en.wikipedia.org/wiki/Coulomb%27s_law

“Wait! VOLTS! I know that one!
240 Volts come out of a wall socket!”

Electricity

- Teeny tiny bit of vector addition
- The components of Coulomb's law parallel to the plates CANCEL, the components perpendicular to the plates ADD, so you get a constant electric field
- Why is this important?
 - If you hook up a battery, this supplies a direct current electric field between the plates

Positive charge



Negative charge

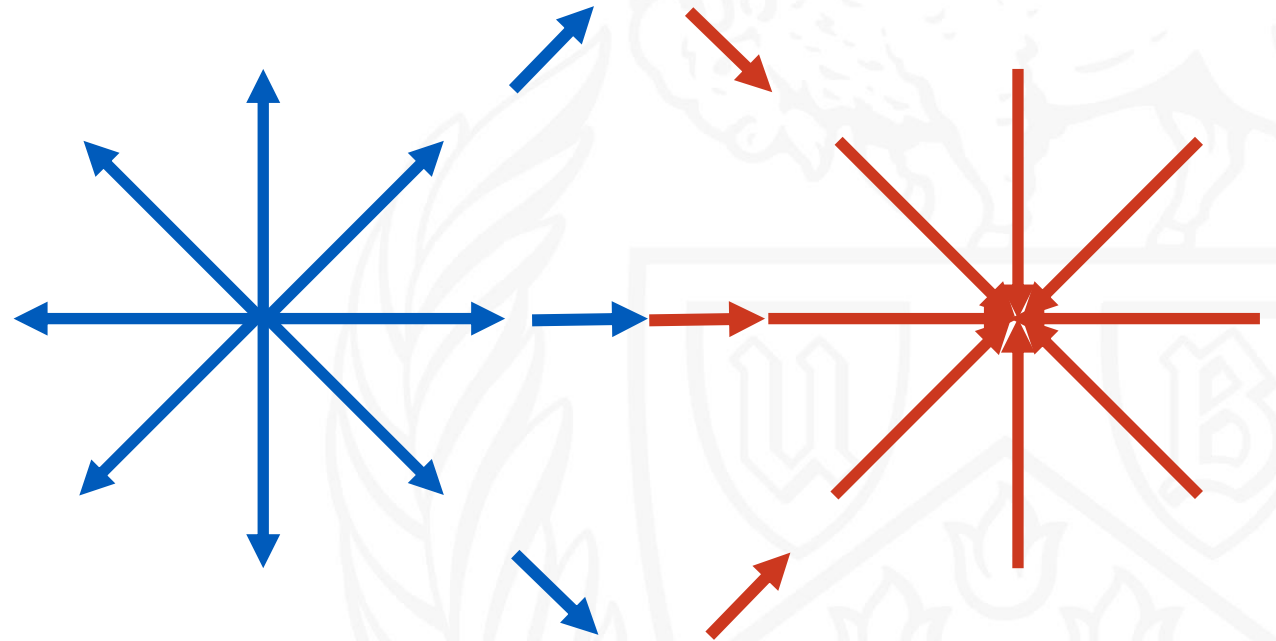


Not responsible for this, just for enrichment

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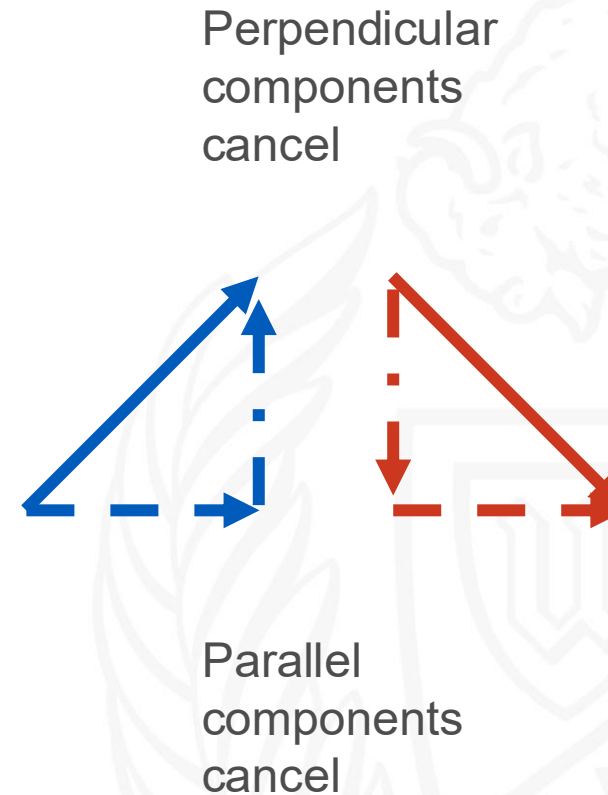
Draw their vector field lines



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Electricity

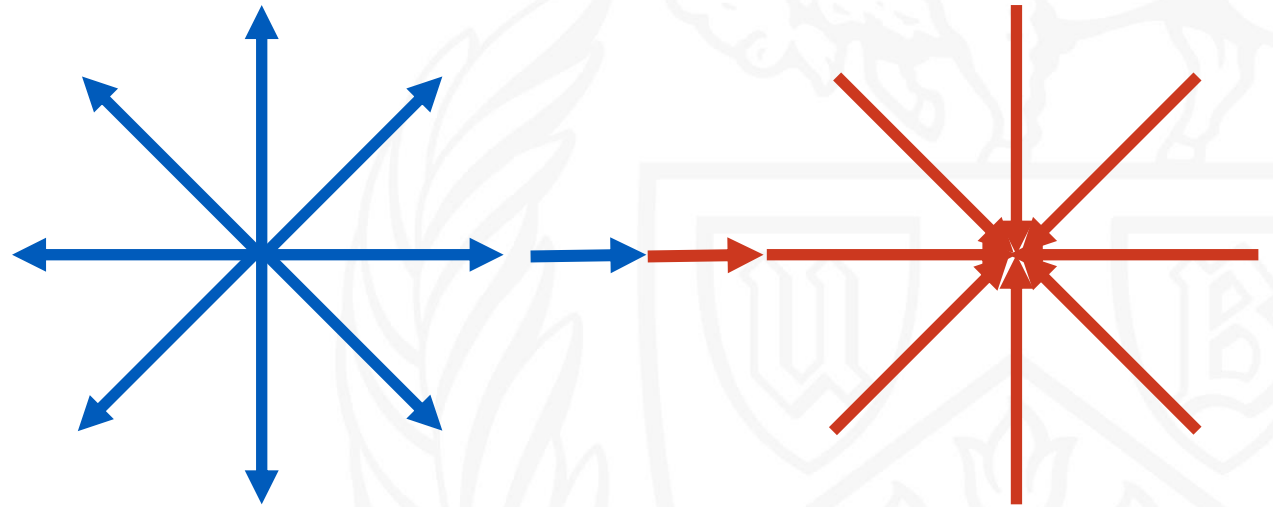
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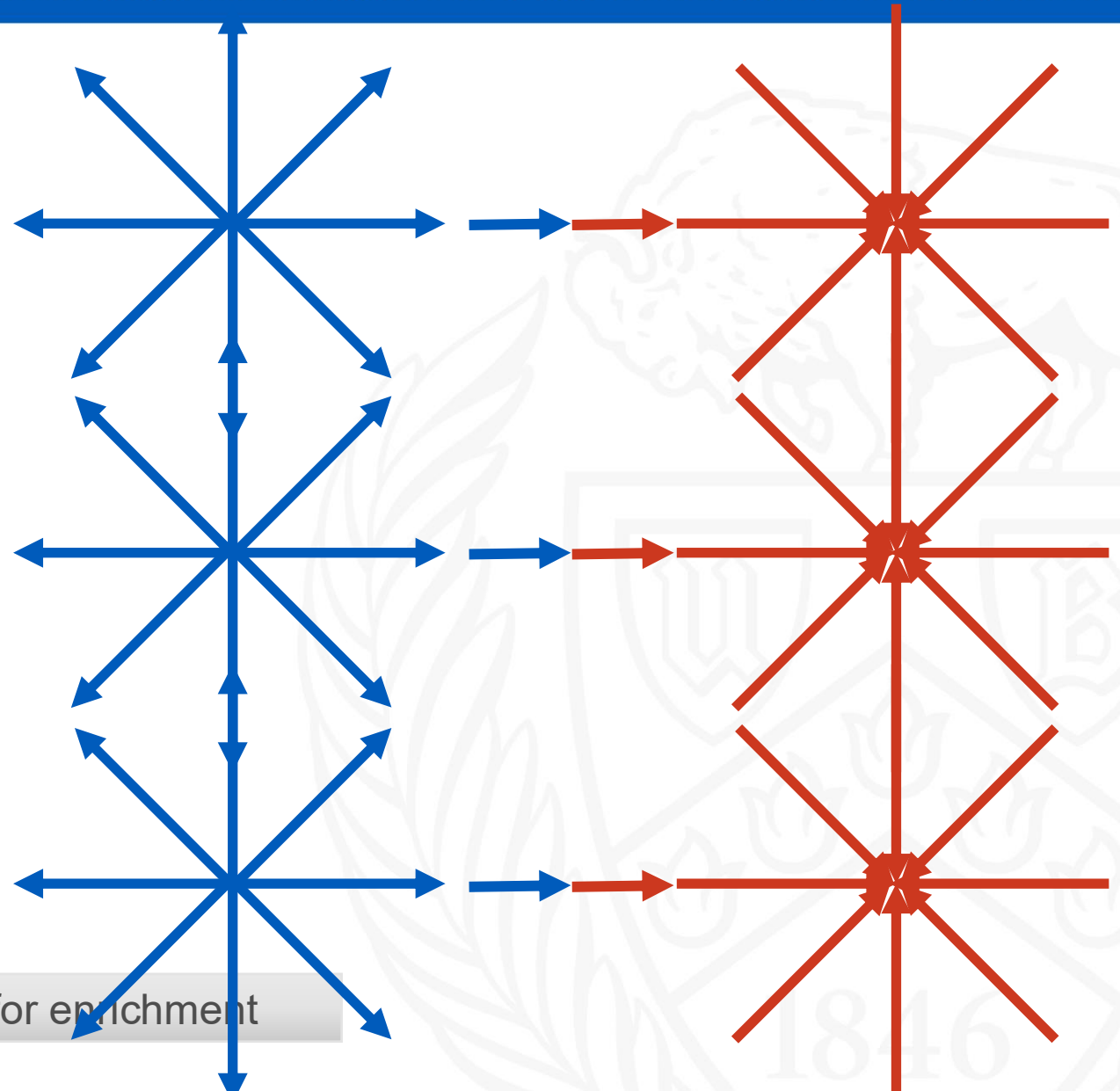
Draw only the part in the middle where it doesn't cancel



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Electricity

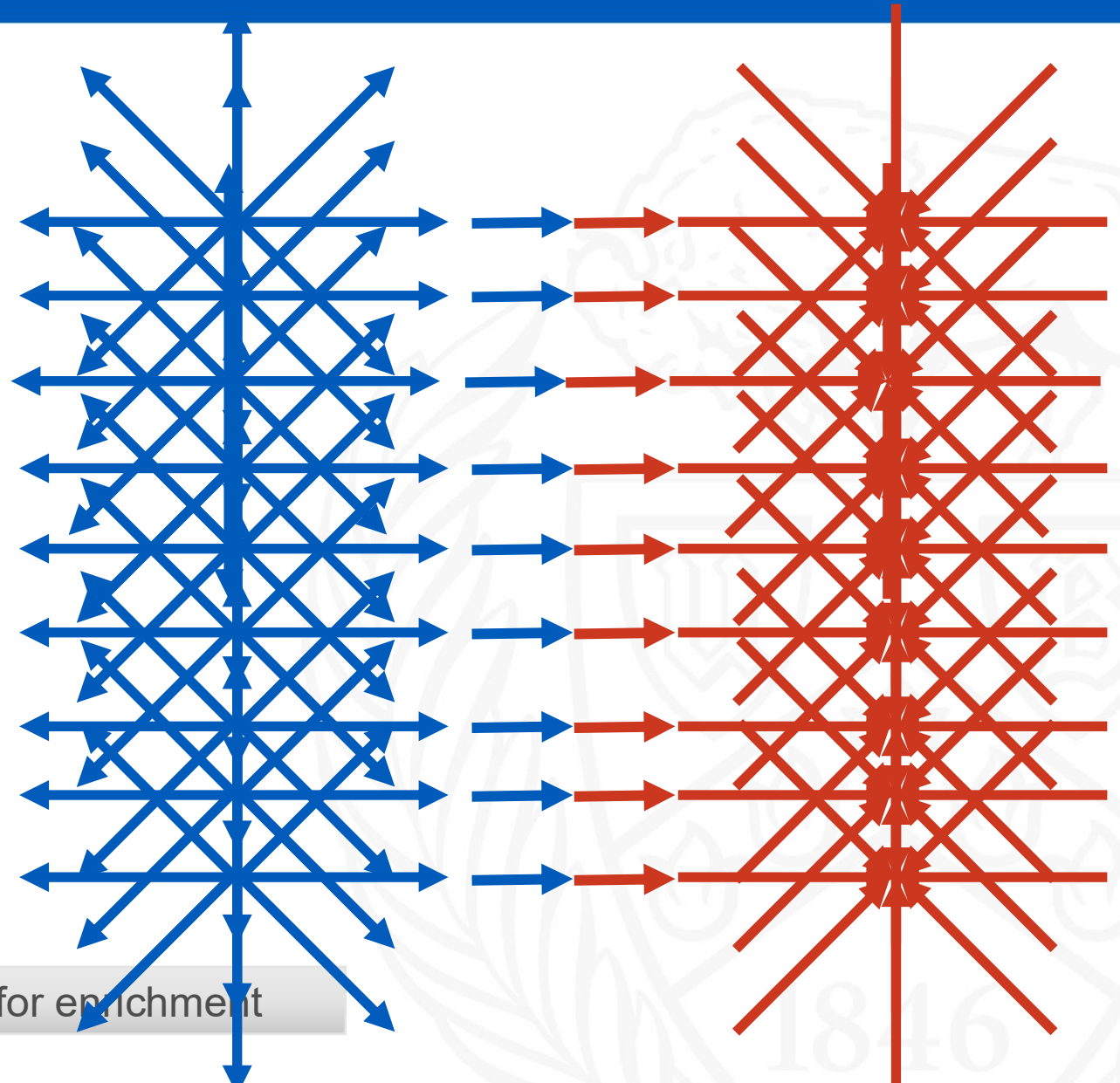
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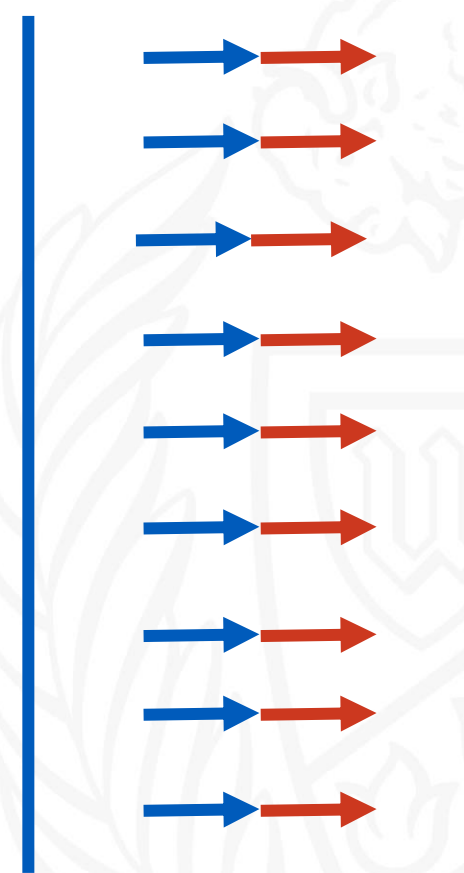


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Now just draw a line to represent the charges, and you get a PARALLEL PLATE CAPACITOR

Electricity

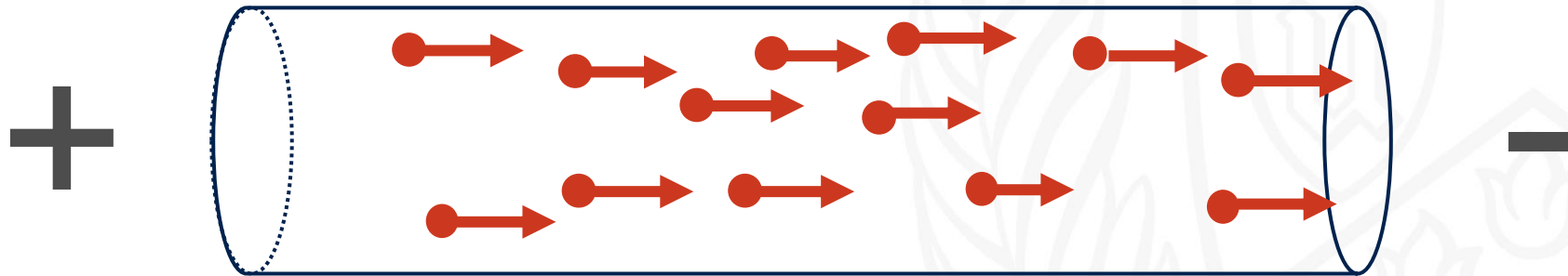
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Electricity

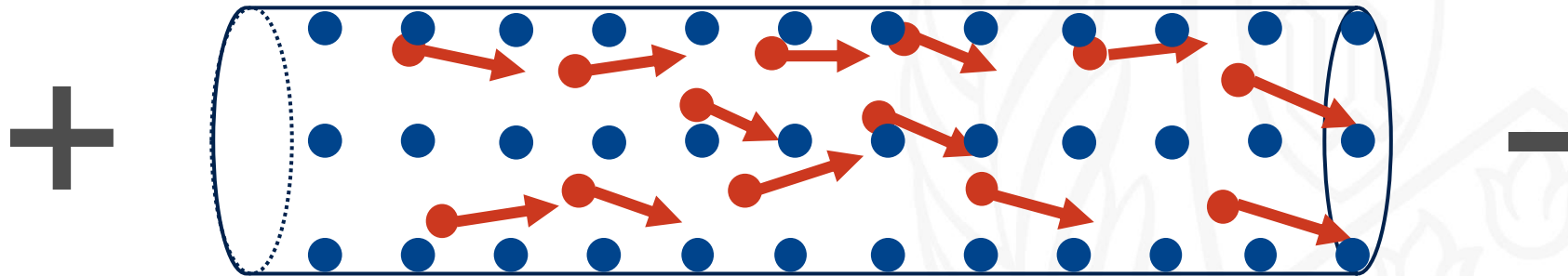
- Then you can attach a WIRE between them, and electrons will be repelled from the negative side and attracted to the positive side



Electricity

- Unfortunately, friction exists
- The electrons flowing through the wires bump into other electrons in the metal, so you get RESISTANCE
- Just like all resistance, it generates heat.

You ARE responsible to understand that resistance is voltage divided by current



Resistance:
Ohms = volts per amp

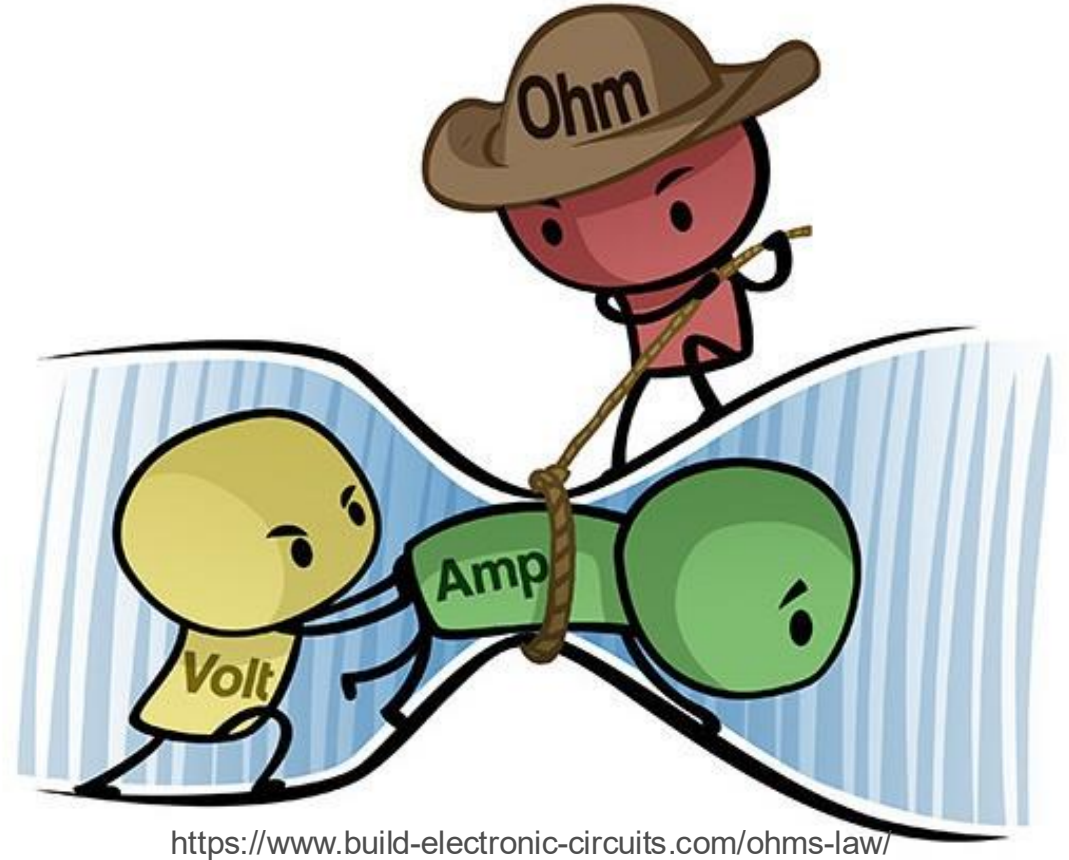
$$R = \frac{V}{i}$$

Electric potential:
Joules / Coulomb or **VOLTS**

Current:
Ampere = Coulombs / second

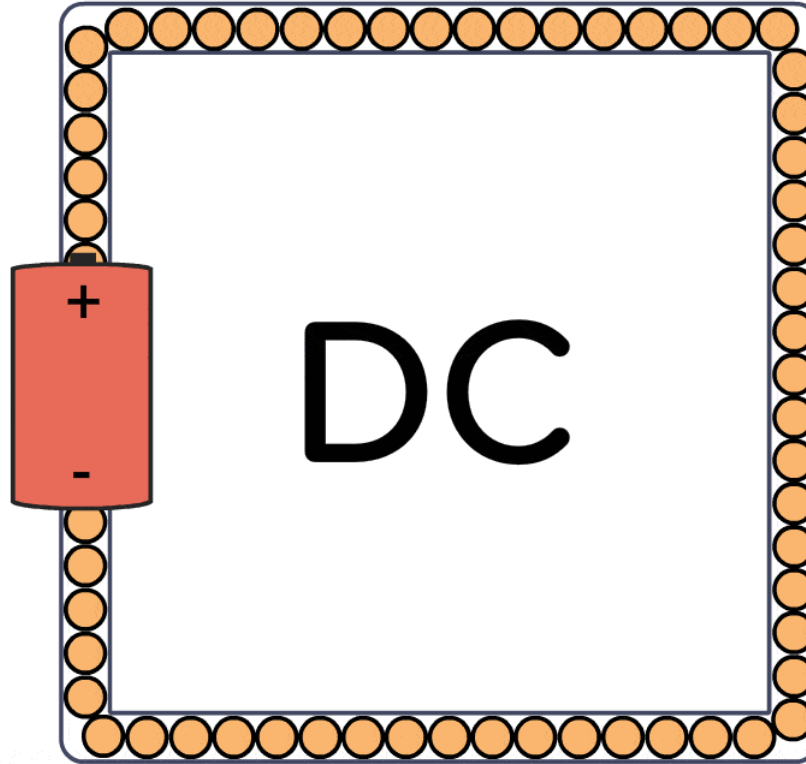
Electricity

- Now we have:
 - **Current** (amps) : charge per unit time
 - **Voltage** (volts) : pushes current
 - **Resistance** (ohms): restricts current



Electricity

- We now have enough knowledge to understand **DIRECT CURRENT (DC)**
- DC is supplied by batteries and solar panels
- You can then combine elements (wires, light bulbs, batteries) to make a **circuit**



<https://www.build-electronic-circuits.com/difference-ac-vs-dc/>

Energy and Power

- Remember that power is energy over time
- But now do a trick... multiply and divide by ΔQ :

$$P = \frac{W}{\Delta t}$$

$$P = \frac{W}{\Delta t} = \frac{W}{\Delta Q} \frac{\Delta Q}{\Delta t}$$

Electric potential:
Joules / Coulomb or **VOLTS**

Electric current:
Coulombs / second or **AMPERES**

$$P = \frac{W}{\Delta t} = V \cdot i$$

You ARE responsible to understand that power in a circuit is the voltage times the current

You ARE responsible to understand these three concepts

Energy and Power

- We're now able to quantify **resistive losses** in a circuit

$$V = iR$$

$$P = iV$$

$$P = i^2 R$$

Critically:

Power dissipated by a circuit is proportional to current times voltage, or current SQUARED times resistance

This sets up the War of the Currents

To remember

- Current is the number of charges per second
- Voltage is the electric energy (or work) divided by the charge
- Power is the product of current and voltage
- Power dissipated in a resistor is current squared times resistance

Concept	Formula
Current is the number of charges per second	$i = \frac{\Delta Q}{\Delta t}$
Voltage is electric energy or work divided by charge	$V = \frac{W}{Q}$
Power is the energy or work per unit time	$P = \frac{W}{\Delta t}$
In a circuit, power is the product of current and voltage	$P = i \cdot V$
In a resistive circuit, power is the product of current squared and resistance	$P = i^2 R$